

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

 show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? Yes

Does R2 show up as an OSPF neighbor on R3? No

What does this information tell you?

All traffic from R3 to the 192.168.2.0/24 network is routed through R1. The S0/0/1 interface on R2 is still configured as passive, preventing OSPF advertisements across that link. The accumulated cost of 129 is calculated from two T1-speed serial links (cost 64 each) plus the Gigabit Ethernet link on R2 (cost 1).

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The cost is 65. This is calculated from the single T1-speed serial link from R3 to R2 (cost 64) plus the Gigabit Ethernet link on R2 (cost 1).

Is R2 listed as an OSPF neighbor to R3? __Yes__

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To achieve a more accurate cost calculation for modern high-speed links (faster than 100 Mb/s). Since the default reference bandwidth is 100 Mb/s, all links faster than this (like Gigabit Ethernet) receive the same cost of 1. Increasing the reference bandwidth provides a larger range of values, allowing for differentiation between Fast Ethernet, Gigabit Ethernet, and 10G interfaces.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24: The cost of R1's S0/0/1 interface (781) added to the cost of R3's G0/0 interface (1), for a total of 782.

Cost to 192.168.23.0/30: The cost of R1's S0/0/1 interface (781) added to the cost of R3's S0/0/1 interface (64), for a total of 845.

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562. Each of the two serial links in the path now has a cost of 781. The route from R1 to the 192.168.23.0/30 network traverses both of these links, so the total accumulated cost is $781 + 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF always selects the path with the lowest accumulated cost. The path via has a total cost of 1563 (781 + 781 + 1). This is lower than the direct path via R3, which now has a manually configured cost of 1566 (1565 + 1). Therefore, the path through R2 is preferred.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling the Router ID ensures stability and predictability in the OSPF process. It prevents the Router ID from changing if a physical interface goes down, which would disrupt OSPF operations. It also allows an administrator to influence the Designated Router (DR) and Backup Designated Router (BDR) election process on multi-access networks.

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election is only relevant on multi-access networks (like Ethernet). The connections between routers in this lab are all point-to-point serial links, which do not require a DR or BDR.

3. Why would you want to set an OSPF interface to passive?

Setting an interface to passive prevents the router from sending OSPF hello packets out of that interface. This is typically done on LAN interfaces to reduce unnecessary routing protocol traffic, enhance security, and conserve bandwidth, while still allowing the connected network to be advertised to other OSPF routers.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas. By dividing a large network into multiple areas, OSPF can limit the propagation of link-state updates and reduce the size of the routing table, which in turn reduces the load on the routers and allows the network to scale.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

~~The route from RIP will be used.~~ Routing decisions are based on Administrative Distance (AD), not the protocol's metric value. RIP has a default AD of 120, OSPF has 110, and EIGRP has 90. Since EIGRP has the lowest AD (most trustworthy), it would normally be chosen. However, if the AD values are the default, the EIGRP route would be selected because its AD of 90 is lower than OSPF's 110. Note: The question's phrasing implies the answer is likely looking for "the route with the lowest metric," which is incorrect. The correct technical answer is the route with the lowest Administrative Distance.

EIGRP

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